The Middle Stair in Parallel Chip-Firing

OpenAI Codex (GPT-5.6 Sol)*
AI research system

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Abstract

Let $G = (V, E)$ be a finite connected simple graph. In parallel chip-firing, every vertex with at least its degree in chips fires simultaneously, sending one chip to each neighbor. We prove that every nonnegative integral configuration σ satisfying

$$2|E| - |V| < |\sigma| < 2|E|$$

has least eventual state period two. This is the statement posed as Conjecture 2 by Ji, Li, and Wang. The main new ingredient is a density inequality: a returning orbit whose common activity is less than one half has at most $2|E| - |V|$ chips. Its proof selects a minimum of a centered firing-prefix function independently at each vertex and pairs the resulting integer correction terms edge by edge. The standard affine chip complement gives the corresponding high-activity bound. These inequalities force activity one half in the strict middle band, after which the nonclumpiness theorem of Jiang, Scully, and Zhang forces alternating firing and hence a two-cycle. The complete theorem and a counterexample to period two from time zero have also been formalized in Lean.

1 Introduction

Parallel chip-firing is a deterministic synchronous dynamical system on integer configurations of a graph. Its eventual behavior has been studied since Bitar and Goles [1]; see also Jiang [3]. On complete graphs, Levine connected the activity function with a devil's staircase and a rotation number [5]. Recent work of Ji, Li, and Wang isolates the middle stair for a general finite connected graph [2]. They pose the following assertion as their Conjecture 2.

Theorem 1.1 (Middle-stair theorem)**.** *Let $G = (V, E)$ be a finite connected simple graph and let $\sigma : V \to \mathbb{Z}_{\geq 0}$ be a chip configuration. If*

$$2|E| - |V| < \sum_{v \in V} \sigma(v) < 2|E|,$$

then the least eventual state period of σ is exactly two.

The theorem was previously known in several graph classes; Ji, Li, and Wang summarize the tree, cycle, complete-graph, and balanced complete-bipartite cases [2]. Our contribution is a graph-independent density inequality for low-activity returning orbits (Lemma 3.1). The complementary

*The documented model slug is `gpt-5.6-sol`. The exact runtime build identifier was not exposed to this session.

inequality follows from the established affine complement for parallel chip-firing [3]. These inequalities force activity 1/2. The remaining step uses the theorem of Jiang, Scully, and Zhang that a periodic firing word cannot contain both 00 and 11 [4, Theorem 6.2].

We make no priority claim beyond the literature review recorded with this draft. Formal verification establishes correctness relative to the stated definitions and dependencies, not originality. The low-activity density inequality is therefore the main point requiring specialist comparison with prior work.

2 Dynamics, recurrence, and common activity

Write $n = |V|$, $m = |E|$, and $d_v = \deg(v)$. A configuration is a map $\sigma : V \to \mathbb{Z}_{\geq 0}$. At time t, define the firing indicator

$$f_t(v) = \begin{cases} 1, & \sigma_t(v) \geq d_v, \\ 0, & \sigma_t(v) < d_v. \end{cases}$$

All active vertices fire simultaneously. If $L = D - A$ is the graph Laplacian, then one round is

$$\sigma_{t+1} = \sigma_t - Lf_t. \tag{1}$$

The total number of chips is conserved, and nonnegative configurations remain nonnegative.

A positive integer p is an *eventual state period* of σ_0 if there is $\mu \geq 0$ such that

$$\sigma_{\mu+p+t} = \sigma_{\mu+t} \qquad (t \geq 0).$$

The least such p is the least eventual state period. This is a period of the state orbit, not merely of an aggregate activity statistic.

Since the coordinates are nonnegative and have fixed sum, every orbit is eventually periodic. Restart at a state on a returning orbit and let $T > 0$ be any return length, so $\sigma_T = \sigma_0$. Put

$$u_v(t) = \sum_{s=0}^{t-1} f_s(v), \qquad 0 \leq t \leq T.$$

Telescoping (1) gives

$$\sigma_t = \sigma_0 - Lu(t). \tag{2}$$

At $t = T$ this yields $Lu(T) = 0$. An integer-valued harmonic function on a connected graph is constant, so there is an integer q, $0 \leq q \leq T$, such that

$$u_v(T) = q \qquad \text{for every } v \in V. \tag{3}$$

Thus every vertex has the same activity q/T. This common-count criterion is standard in the subject; compare [1, 3].

We will use the following established firing-word theorem. Regard $(f_0(v), \ldots, f_{T-1}(v))$ as a cyclic binary word.

Theorem 2.1 (Jiang–Scully–Zhang nonclumpiness)**.** *On a returning parallel chip-firing orbit, the cyclic firing word of every vertex does not contain both a cyclic occurrence of* 00 *and a cyclic occurrence of* 11*.*

This is the nonclumpiness theorem of Jiang, Scully, and Zhang [4, Theorem 6.2]. It enters only after the density argument has forced $q/T = 1/2$.

3 A low-activity density inequality

Lemma 3.1 (Low-activity density bound). *Let $\sigma_0, \ldots, \sigma_T = \sigma_0$ be a returning orbit on a finite connected simple graph. Let q be the common firing count in (3). If $2q < T$, then*

$$|\sigma_0| := \sum_{v \in V} \sigma_0(v) \le 2m - n.$$

Proof. For each vertex v, define the denominator-cleared centered prefix height

$$h_v(t) = T u_v(t) - qt, \qquad 0 \le t \le T.$$

Equation (3) gives $h_v(T) = h_v(0) = 0$, so h_v may be viewed on the cyclic set of phases modulo T. Independently for every vertex, select a phase at which h_v is minimal. Let $\tau_v \in \{0, \ldots, T-1\}$ be the round immediately preceding that phase. When $\tau_v = T-1$, the endpoint T represents phase zero.

Minimality against the preceding phase gives

$$0 \ge h_v(\tau_v + 1) - h_v(\tau_v) = T f_{\tau_v}(v) - q.$$

Because $q < T$ and $f_{\tau_v}(v) \in \{0, 1\}$, this forces

$$f_{\tau_v}(v) = 0. \tag{4}$$

Fix an edge $\{v, w\}$. The independently selected minima satisfy

$$h_v(\tau_v + 1) \le h_v(\tau_w), \qquad h_w(\tau_w + 1) \le h_w(\tau_v).$$

Using (4) and adding these inequalities gives

$$T K_{vw} \le 2q, \tag{5}$$

where

$$K_{vw} = u_v(\tau_v) - u_w(\tau_v) + u_w(\tau_w) - u_v(\tau_w).$$

The number K_{vw} is an integer. Since $2q < T$, inequality (5) implies

$$K_{vw} \le 0. \tag{6}$$

Vertex v waits at time τ_v, so chip integrality gives $\sigma_{\tau_v}(v) \le d_v - 1$. Applying (2) at the vertex-dependent time τ_v yields

$$\sigma_0(v) \le d_v - 1 + (Lu(\tau_v))_v.$$

Every one of these inequalities bounds a coordinate of the same initial configuration, so they may be summed even though the selected times differ. Regrouping the Laplacian terms by unoriented edges gives

$$\begin{aligned} |\sigma_0| &\le \sum_{v \in V} (d_v - 1) + \sum_{\{v,w\} \in E} K_{vw} \\ &\le 2m - n, \end{aligned}$$

where the last inequality uses (6). This proves the claim. □

The bound is attained, for every connected graph with at least two vertices, by the fixed configuration $\sigma(v) = d_v - 1$. Thus its numerical threshold cannot be improved without further hypotheses.

4 Complementation and the high-activity bound

Lemma 4.1 (High-activity density bound)**.** *Let $\sigma_0, \ldots, \sigma_T = \sigma_0$ be a returning orbit with common firing count q. If $T/2 < q < T$, then*
$$|\sigma_0| \geq 2m.$$

Proof. Since $q < T$, every vertex waits at least once during the return interval. At a time when v waits, its chip count is at most $d_v - 1$. The coordinatewise upper bound

$$\sigma_t(v) \leq 2d_v - 1 \tag{7}$$

then persists under one update: a waiting vertex receives at most d_v chips, while a firing vertex loses d_v chips and receives at most d_v. Following the returning orbit once from a waiting phase for each v proves (7) at every phase.

Define the affine complement
$$\sigma_t^c(v) = 2d_v - 1 - \sigma_t(v).$$
It is nonnegative by (7). The threshold rule gives $f_t^c = \mathbf{1} - f_t$, and $L\mathbf{1} = 0$ shows that complementation commutes with the update. Hence σ^c is a returning orbit with common firing count $T - q < T/2$. Lemma 3.1 gives $|\sigma_0^c| \leq 2m - n$. On the other hand,

$$|\sigma_0^c| = \sum_{v \in V} (2d_v - 1) - |\sigma_0| = 4m - n - |\sigma_0|.$$

Rearrangement gives $|\sigma_0| \geq 2m$. ∎

The configuration $\sigma(v) = d_v$ is fixed because every vertex fires and $L\mathbf{1} = 0$; it attains the threshold $2m$. Together with the example following Lemma 3.1, this explains why the inequalities in Theorem 1.1 must be strict.

5 Proof of the middle-stair theorem

Proof of Theorem 1.1. Restart the eventual orbit at a returning state, and let $T > 0$ be a return length with common firing count q. The chip total remains in the strict middle band.

The cases $q = 0$ and $q = T$ are impossible. In the first case all vertices wait throughout the return interval, so $\sigma(v) \leq d_v - 1$ and $|\sigma| \leq 2m - n$. In the second case all vertices fire, so $\sigma(v) \geq d_v$ and $|\sigma| \geq 2m$.

If $2q < T$, Lemma 3.1 contradicts the lower strict inequality. If $T < 2q$, then $q < T$ and Lemma 4.1 contradicts the upper strict inequality. Therefore

$$2q = T. \tag{8}$$

Each cyclic firing word has q ones and $T - q = q$ zeros. By Theorem 2.1, it has no cyclic 00 or no cyclic 11. A balanced cyclic binary word with either property must alternate. Consequently

$$f_{t+1}(v) = 1 - f_t(v)$$

for every vertex and every phase on the returning orbit. Applying (1) twice now gives

$$\begin{aligned}\sigma_{t+2} &= \sigma_t - L(f_t + f_{t+1}) \\ &= \sigma_t - L\mathbf{1} = \sigma_t.\end{aligned}$$

Thus two is an eventual state period.

It remains to exclude eventual period one. At a fixed state, $Lf = 0$; by connectedness, the binary firing indicator f is constant. If $f = 0$, then $|\sigma| \le 2m - n$, and if $f = \mathbf{1}$, then $|\sigma| \ge 2m$. Conservation of the chip total contradicts the strict band in either case. Hence one is not an eventual period, and the least eventual state period is exactly two. $\square$

6 Eventuality and boundary cases

The word "eventual" in Theorem 1.1 cannot be removed. Let G be the path on three vertices, with the first coordinate at the center, and start from

$$(0, 0, 2).$$

Here $n = 3$, $m = 2$, and $1 < 2 < 4$. Direct application of the parallel update gives

$$(0, 0, 2) \longmapsto (1, 0, 1) \longmapsto (2, 0, 0) \longmapsto (0, 1, 1) \longmapsto (2, 0, 0).$$

Thus the initial state does not return after two rounds; the two-cycle begins at time two. The displayed transitions and both period claims are checked in the accompanying Lean development.

The endpoint configurations $d_v - 1$ and d_v are fixed and have totals $2m - n$ and $2m$, respectively, when the connected graph has at least two vertices. Consequently neither strict inequality in the theorem can be replaced by a weak inequality. For the one-vertex connected graph, the strict middle band contains no nonnegative chip total.

7 Formal verification

The accompanying Lean 4 development uses Lean 4.31.0 and mathlib v4.31.0. Its terminal declaration is

`MiddleStair.middle_stair_least_eventual_period_two`.

The definitions distinguish returning cycles, eventual state periods, and least positive eventual state periods. The development formalizes finite recurrence, common firing counts, Lemmas 3.1 and 4.1, half activity, alternation, the two-step return, and the exclusion of eventual period one. It also formalizes the three-vertex counterexample above.

For Theorem 2.1, the development formalizes the sector and mischief argument of Jiang, Scully, and Zhang. Its only exhaustive local step checks a 256^2-pair reduced-cost certificate by Lean kernel reduction; the subsequent graph-level argument is symbolic. This provides a closed formal dependency, but fidelity of the encoding to the published proof remains a human-review question distinct from kernel correctness.

The project contains no `sorry`, `admit`, or declared project axiom. An included `#print axioms` audit reports only Lean and mathlib logical foundations. Exact build instructions and a proof-to-code map are included in the handoff repository.

Authorship and disclosure

At Nathan Lannan's explicit direction, OpenAI Codex (GPT-5.6 Sol) is listed as first author and Nathan Lannan as second author. The AI system materially assisted with literature triage, proof exploration, Lean formalization, adversarial checking, and manuscript drafting. This attribution is nontraditional and may not satisfy a journal's authorship policy. Nathan Lannan has no institutional affiliation and can be contacted at nathan@nathanlannan.com. He must approve the final text and accept the human scholarly responsibility required before submission.

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